

A9 Express – GitHub Backlog (20 Professional Issues)

Project: A9 Express – Unified Transport Booking Platform

Repository: `documents` / `research` / `prototypes`

Sprint Structure based on SENG 31242 Guidelines.

Issue #1 – Create GitHub Organisation and Repository Structure

Labels: epic:project-setup, P1-Blocker, task

Sprint: Sprint 1

User Story

As a project team, we want to create the GitHub organisation and repositories so that all project artefacts are managed centrally.

Acceptance Criteria

- ☐ GitHub organisation created using naming convention.
- ☐ `documents`, `research`, and `prototypes` repositories created.
- ☐ All team members added.
- ☐ Supervisors added as outside collaborators.
- ☐ README.md added to all repositories.

Tasks

- ☐ Create organisation.
- ☐ Create repositories.
- ☐ Configure permissions.
- ☐ Add README.md and .gitignore.

Estimate

Estimated: 3 hours

Issue #2 – Configure Branch Protection Rules

Labels: epic:project-setup, P2-Critical, task

Sprint: Sprint 1

User Story

As a team member, I want protected branches so that accidental changes to the main branch are prevented.

Acceptance Criteria

- ☐ PR review required before merge.
- ☐ Linear history enabled.
- ☐ Stale approvals dismissed.
- ☐ Administrators included.

Tasks

- ☐ Open repository settings.
- ☐ Configure branch protection.
- ☐ Test pull request workflow.

Estimate

Estimated: 2 hours

Issue #3 – Create GitHub Project Board

Labels: epic:project-setup, P2-Critical, task

Sprint: Sprint 1

User Story

As a project team, we want a GitHub Kanban board so that project tasks can be tracked efficiently.

Acceptance Criteria

- ☐ Project board created.
- ☐ Columns configured.
- ☐ Custom fields added.
- ☐ Sprint milestones linked.

Tasks

- ☐ Create board.
- ☐ Add Backlog, To Do, In Progress, In Review, Done columns.
- ☐ Configure priority and sprint fields.

Estimate

Estimated: 3 hours

Issue #4 – Conduct Competitive Analysis

Labels: epic:fact-finding, P2-Critical, research

Sprint: Sprint 1

User Story

As a business analyst, I want to compare existing transport booking platforms so that system gaps can be identified.

Acceptance Criteria

- ☐ At least 3 existing solutions analysed.
- ☐ Feature comparison matrix completed.
- ☐ Pain points documented.
- ☐ Findings committed to research repository.

Tasks

- ☐ Research SLTB booking systems.
- ☐ Analyse BusSeat.lk and competitors.
- ☐ Prepare comparison chart.

Estimate

Estimated: 5 hours

Issue #5 – Conduct Client Interview Session 01

Labels: epic:fact-finding, P1-Blocker, research

Sprint: Sprint 2

User Story

As a requirements engineer, I want to conduct a structured client interview so that accurate business requirements can be collected.

Acceptance Criteria

- ☐ Interview questions prepared.
- ☐ Client interview conducted.
- ☐ Meeting notes documented.
- ☐ Key findings summarised.

Tasks

- ☐ Prepare questionnaire.
- ☐ Conduct interview.
- ☐ Upload meeting notes.

Estimate

Estimated: 4 hours

Issue #6 – Identify System Actors and Personas

Labels: epic:requirements, P3-Major, documentation

Sprint: Sprint 2

User Story

As a system analyst, I want to identify system actors and personas so that user requirements are clearly understood.

Acceptance Criteria

- ☐ All primary actors identified.
- ☐ User personas documented.
- ☐ Stakeholder map completed.

Tasks

- ☐ Define passengers.
- ☐ Define operators.
- ☐ Define administrators.
- ☐ Create persona documents.

Estimate

Estimated: 4 hours

Issue #7 – Create System Context Diagram

Labels: epic:requirements, P2-Critical, documentation

Sprint: Sprint 2

User Story

As a system analyst, I want a context diagram so that the external system interactions are visualised.

Acceptance Criteria

- ☐ Context diagram created.
- ☐ External entities identified.
- ☐ Diagram exported as SVG.
- ☐ Source file committed.

Tasks

- ☐ Draw context diagram.
- ☐ Validate system boundaries.
- ☐ Export diagram.

Estimate

Estimated: 3 hours

Issue #8 – Create Use Case Diagram

Labels: epic:requirements, P1-Blocker, documentation

Sprint: Sprint 2

User Story

As a requirements engineer, I want a use case diagram so that all system functionalities are represented visually.

Acceptance Criteria

- ☐ Use case diagram completed.
- ☐ All actors included.
- ☐ UML notation followed.
- ☐ Source and export files committed.

Tasks

- ☐ Identify use cases.
- ☐ Draw UML diagram.
- ☐ Review with team.

Estimate

Estimated: 5 hours

Issue #9 – Draft Use Case UC-01 User Registration

Labels: epic:requirements, P2-Critical, documentation

Sprint: Sprint 2

User Story

As a new user, I want to register an account so that I can access booking services.

Acceptance Criteria

- ☐ Use case follows SRS template.
- ☐ Main flow defined.
- ☐ Alternative flows included.
- ☐ Business rules documented.

Tasks

- ☐ Write UC-01.
- ☐ Review use case.
- ☐ Commit markdown file.

Estimate

Estimated: 3 hours

Issue #10 – Draft Use Case UC-02 Bus Ticket Booking

Labels: epic:requirements, P1-Blocker, documentation

Sprint: Sprint 2

User Story

As a passenger, I want to reserve bus tickets online so that I can travel conveniently.

Acceptance Criteria

- ☐ Booking flow documented.
- ☐ Seat selection included.
- ☐ Payment process included.
- ☐ Alternative flows documented.

Tasks

- ☐ Write use case.
- ☐ Add business rules.
- ☐ Peer review.

Estimate

Estimated: 5 hours

Issue #11 – Draft Use Case UC-03 Airport Transfer Booking

Labels: epic:requirements, P2-Critical, documentation

Sprint: Sprint 2

User Story

As a traveller, I want to book airport transport so that I can schedule reliable pickups and drop-offs.

Acceptance Criteria

- ☐ Pickup scheduling documented.
- ☐ Driver allocation included.
- ☐ Booking confirmation defined.

Tasks

- ☐ Prepare use case.
- ☐ Validate flows.
- ☐ Commit files.

Estimate

Estimated: 4 hours

Issue #12 – Draft Use Case UC-04 Tour Package Booking

Labels: epic:requirements, P2-Critical, documentation

Sprint: Sprint 2

User Story

As a customer, I want to reserve custom tour packages so that I can plan trips easily.

Acceptance Criteria

- ☐ Tour customization included.
- ☐ Pricing flow documented.
- ☐ Booking confirmation defined.

Tasks

- ☐ Write use case.
- ☐ Add alternative flows.
- ☐ Review with team.

Estimate

Estimated: 4 hours

Issue #13 – Create Activity Diagram for Ticket Booking

Labels: epic:requirements, P3-Major, documentation

Sprint: Sprint 2

User Story

As a designer, I want an activity diagram for ticket booking so that the workflow is visualised clearly.

Acceptance Criteria

- ☐ UML activity diagram completed.
- ☐ Decision points included.
- ☐ SVG export committed.

Tasks

- ☐ Design workflow.
- ☐ Create diagram.
- ☐ Export files.

Estimate

Estimated: 3 hours

Issue #14 – Write Functional Requirements Section

Labels: epic:requirements, P1-Blocker, documentation

Sprint: Sprint 3

User Story

As a documentation lead, I want to define functional requirements so that system features are formally specified.

Acceptance Criteria

- ☐ FR-01 to FR-15 documented.
- ☐ Priority defined.
- ☐ Verification method added.
- ☐ Sources referenced.

Tasks

- ☐ Draft requirements.
- ☐ Validate with use cases.
- ☐ Peer review.

Estimate

Estimated: 6 hours

Issue #15 – Write Non-Functional Requirements

Labels: epic:requirements, P2-Critical, documentation

Sprint: Sprint 3

User Story

As a system architect, I want non-functional requirements documented so that quality standards are defined.

Acceptance Criteria

- ☐ Security requirements documented.
- ☐ Performance requirements documented.
- ☐ Scalability requirements documented.
- ☐ Reliability requirements documented.

Tasks

- ☐ Gather NFRs.
- ☐ Define measurable targets.
- ☐ Review with team.

Estimate

Estimated: 5 hours

Issue #16 – Select System Architecture Pattern

Labels: epic:architecture, P1-Blocker, design

Sprint: Sprint 3

User Story

As a software architect, I want to select a suitable architecture pattern so that the system is scalable and maintainable.

Acceptance Criteria

- ☐ Architecture style selected.
- ☐ ADR document created.
- ☐ Alternatives analysed.
- ☐ Trade-offs documented.

Tasks

- ☐ Compare MVC and microservices.
- ☐ Create ADR.
- ☐ Conduct review meeting.

Estimate

Estimated: 5 hours

Issue #17 – Create High-Level Architecture Diagram

Labels: epic:architecture, P2-Critical, design

Sprint: Sprint 3

User Story

As a software architect, I want a high-level architecture diagram so that system components are clearly represented.

Acceptance Criteria

- ☐ Frontend, backend, and database components shown.
- ☐ External services included.
- ☐ Deployment relationships shown.

Tasks

- ☐ Design architecture.
- ☐ Create UML diagram.
- ☐ Export SVG.

Estimate

Estimated: 4 hours

Issue #18 – Design Database ER Diagram

Labels: epic:class-design, P1-Blocker, design

Sprint: Sprint 3

User Story

As a database designer, I want an ER diagram so that data relationships are properly structured.

Acceptance Criteria

- ☐ Entities identified.
- ☐ Relationships defined.
- ☐ Primary and foreign keys included.
- ☐ Diagram reviewed.

Tasks

- ☐ Define tables.
- ☐ Create ER diagram.
- ☐ Validate normalization.

Estimate

Estimated: 5 hours

Issue #19 – Design UI Wireframes

Labels: epic:ui-design, P2-Critical, design

Sprint: Sprint 4

User Story

As a UX designer, I want wireframes for major screens so that the user interface is planned effectively.

Acceptance Criteria

- ☐ Home page wireframe completed.
- ☐ Booking screens designed.
- ☐ Dashboard screens designed.
- ☐ Mobile responsiveness considered.

Tasks

- ☐ Create Figma designs.
- ☐ Export screenshots.
- ☐ Conduct design review.

Estimate

Estimated: 6 hours

Issue #20 – Finalise SRS and SDS Documentation

Labels: epic:documentation, P1-Blocker, documentation

Sprint: Sprint 4

User Story

As a project team, we want final SRS and SDS documents completed so that the project is ready for submission.

Acceptance Criteria

- ☐ SRS completed.
- ☐ SDS completed.
- ☐ Grammar and formatting checked.
- ☐ PDF exports generated.
- ☐ Documents committed to repository.

Tasks

- ☐ Proofread documents.
- ☐ Verify references.
- ☐ Export final PDFs.
- ☐ Create final pull request.

Estimate

Estimated: 8 hours